

● HARD

● INFORMATION AND IDEAS

Central Ideas and Details

10 questions · ~15 min

02

Q1

INFORMATION AND IDEAS

The following text is adapted from Lewis Carroll's 1889 satirical novel *Sylvie and Bruno*. A crowd has gathered outside a room belonging to the Warden, an official who reports to the Lord Chancellor.

One man, who was more excited than the rest, flung his hat high into the air, and shouted (as well as I could make out) "Who roar for the Sub-Warden?" Everybody roared, but whether it was for the Sub-Warden, or not, did not clearly appear: some were shouting "Bread!" and some "Taxes!", but no one seemed to know what it was they really wanted.

All this I saw from the open window of the Warden's breakfast-saloon, looking across the shoulder of the Lord Chancellor.

"What can it all mean?" he kept repeating to himself. "I never heard such shouting before—and at this time of the morning, too! And with such unanimity!"

Based on the text, how does the Lord Chancellor respond to the crowd?

- A. He asks about the meaning of the crowd's shouting, even though he claims to know what the crowd wants.
- B. He indicates a desire to speak to the crowd, even though the crowd has asked to speak to the Sub-Warden.
- C. He expresses sympathy for the crowd's demands, even though the crowd's shouting annoys him.
- D. He describes the crowd as being united, even though the crowd clearly appears otherwise.

Jürgen Kocka and other historians of capitalism rarely discuss domestic capitalism in Africa before the period of European colonization, implicitly presenting capitalism as external to and imposed on Africa. Crislayne Alfagali and other Africanist scholars have shown, however, that in parts of Africa, returns-focused investment, the establishment of open markets for wage labor, and other features of capitalism predated colonization. One reason for this discrepancy is that historians of capitalism tend to focus on longitudinal economic data drawn from archival records, which do not exist for much of precolonial Africa.

Which statement about Alfagali and other Africanist scholars is best supported by information in the text?

- A. They likely make use of different types of evidence than historians of capitalism typically rely on.
- B. They likely differ from historians of capitalism in the methods they use to derive longitudinal economic data from archival records.
- C. They likely have a different view about which activities should be considered capitalist in nature than historians of capitalism do.
- D. They likely view capitalism as having been more beneficial for Africa than historians of capitalism do.

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

- A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.
- B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.
- C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.
- D. Before 2023, researchers' inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.

The following text is from Thomas Mann's 1924 novel *The Magic Mountain*, translated by John E. Woods in 1995.

The story of Hans Castorp that we intend to tell here—not for his sake (for the reader will come to know him as a perfectly ordinary, if engaging young man), but for the sake of the story itself, which seems to us to be very much worth telling (although in Hans Castorp's favor it should be noted that it is *his* story, and that not every story happens to everybody)—is a story that took place long ago, and is, so to speak, covered with the patina of history and must necessarily be told with verbs whose tense is that of the deepest past.

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What does the text most strongly suggest about the story of Hans Castorp?

- A. Though it is true that stories of even the most uninteresting people are themselves interesting because all people are unique, the reason this story is interesting is nonetheless difficult to understand because of the passage of time.
- B. Even though it is a story of a person of no particular importance, its age and the manner in which it therefore must be told are both indicators that the story itself is important.
- C. Like all stories about the lives of inconsequential people, this story must necessarily be related in a particular way if the reason the story is consequential is to be made evident to the audience.
- D. It is a remarkable story that happened to an unremarkable person, though one could plausibly argue that because the story is valuable, some of its value accrues to the person at its center.

To understand how temperature change affects microorganism-mediated cycling of soil nutrients in alpine ecosystems, Eva Kaštovská et al. collected plant-soil cores in the Tatra Mountains at elevations around 2,100 meters and transplanted them to elevations of 1,700–1,800 meters, where the mean air temperature was warmer by 2°C. Microorganism-mediated nutrient cycling was accelerated in the transplanted cores; crucially, microorganism community composition was unchanged, allowing Kaštovská et al. to attribute the acceleration to temperature-induced increases in microorganism activity.

It can most reasonably be inferred from the text that the finding about the microorganism community composition was important for which reason?

- A. It provided preliminary evidence that microorganism-mediated nutrient cycling was accelerated in the transplanted cores.
- B. It suggested that temperature-induced changes in microorganism activity may be occurring at increasingly high elevations.
- C. It ruled out a potential alternative explanation for the acceleration in microorganism-mediated nutrient cycling.
- D. It clarified that microorganism activity levels in the plant-soil cores varied depending on which microorganisms comprised the community.

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell's electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as “highly reproducible, concise, and practical.”

What does the text most strongly suggest about conventional ETL fabrication?

- A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.
- B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.
- C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.
- D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

The following text is from Maggie Pogue Johnson's 1910 poem "Poet of Our Race." In this poem, the speaker is addressing Paul Laurence Dunbar, a Black author.

Thou, with stroke of mighty pen,
 Hast told of joy and mirth,
And read the hearts and souls of men
 As cradled from their birth.

The language of the flowers,
 Thou hast read them all,
And e'en the little brook
 Responded to thy call.

Which choice best states the main purpose of the text?

- A. To praise a certain writer for being especially perceptive regarding people and nature
- B. To establish that a certain writer has read extensively about a variety of topics
- C. To call attention to a certain writer's careful and elaborately detailed writing process
- D. To recount fond memories of an afternoon spent in nature with a certain writer

Artificial leaves are a developing renewable energy technology that mimics the process of photosynthesis in plants. These devices are silicon-based solar cells coated in chemical catalysts that activate reactions that split water molecules into hydrogen and oxygen gas. The technology, while generating lots of interest, is not yet commercially viable as a large-scale energy source. To meet this challenge, scientists from many fields are researching ways to store, transport, and distribute the energy the devices produce while other scientists are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

- A. Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an energy source.
- B. The recent increase in the commercial use of artificial leaves as an energy source has encouraged many scientists to research ways to improve the technology.
- C. Artificial leaves split water molecules into oxygen and hydrogen gas using catalysts more efficiently than plants do using the process of photosynthesis.
- D. Artificial leaves were developed to mimic the natural process of photosynthesis in plants in order to store energy for long-term commercial use.

In superfluorescence, electrical charges known as dipoles emit light in synchronized bursts so intense that they are visible to the eye. Until recently, this phenomenon has only been observed at extremely cold temperatures because dipoles cannot synchronize at higher temperatures. But in a study, Melike Biliroglu and colleagues observed superfluorescence at room temperature in thin films made of perovskite and other similarly crystalline materials; the researchers propose that the formation of shock-absorbing quasiparticles called polarons in the material protects dipoles from thermal interference.

Based on the text, how are polarons believed to be involved in the superfluorescence observed in Biliroglu and colleagues' study?

- A. Polarons enable superfluorescent bursts to cross from one crystalline material to another.
- B. Polarons allow for the dipoles to synchronize despite higher temperatures.
- C. Polarons accelerate the dipoles' release of superfluorescent bursts.
- D. Polarons decrease the intensity of the superfluorescent burst.

Elizabeth Asiedu has identified a negative correlation between the share of developing countries' economies derived from natural-resource extraction and those countries' receipts of foreign investment. This may appear counterintuitive—resource extraction requires initial investments (in extractive technology, for instance) at scales best met by multinational corporations—but Asiedu notes that natural-resource industries' boom-bust cycle can destabilize local currencies and increase developing countries' vulnerability to external shocks, creating levels of uncertainty to which foreign investors are typically averse.

Which choice best states the main idea of the text?

- A. Although it may seem surprising that foreign investment declines in developing countries as natural-resource extraction makes up a larger share of those countries' economies, that decline happens because resource extraction requires initial investments too large for foreign investors to supply.
- B. Although developing countries tend to become less dependent on foreign investment as natural-resource industries make up a larger share of their economies, this change may not occur if the boom-bust cycle of those industries destabilizes local currencies or increases countries' vulnerability to external shocks.
- C. Although one might expect that foreign investment would increase as natural-resource extraction makes up a larger share of developing countries' economies, the opposite happens because heavy reliance on natural resources can lead to unattractive conditions for investors.
- D. Although foreign investors tend to avoid initial investments in natural-resource industries in developing countries, foreign investment may increase significantly as those industries stabilize and the risks associated with them decline.